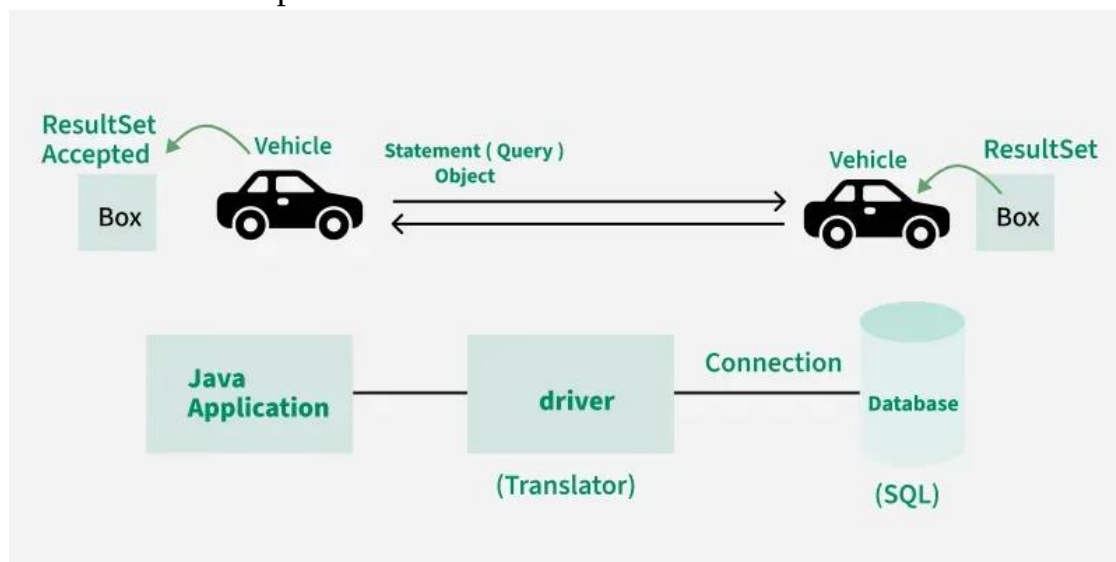


JDBC

- JDBC is an API, meaning it's a set of interfaces and classes that define how Java code can interact with databases.
- JDBC allows Java applications to connect to and work with different databases without needing to rewrite the code for each specific database. This is achieved through JDBC drivers.
- JDBC is a standard Java API, ensuring consistency and portability across different Java environments and database systems.
- JDBC provides the necessary classes and methods to establish connections to various databases.
- You can send SQL queries and commands to the database using JDBC.
- JDBC allows you to retrieve the results of your queries and process them within your Java application.
- In essence, JDBC acts as a bridge between Java applications and databases, enabling seamless interaction and data manipulation.

Steps to Establish a JDBC Connection

- The diagram below demonstrates the workings of JDBC by correlating its steps to real-world examples.



- Below are the steps that explain how to connect to Database in Java:

Step 1: Import the Packages

Step 2: Load the drivers using the `forName()` method

Step 3: Establish a connection using the Connection class object

Step 4: Create a statement

Step 5: Execute the query

Step 6: Close the connections

Step-1: Import JDBC Packages

- The Import statements tell the Java compiler where to find the classes you reference in your code and are placed at the very beginning of your source code.
- To use the standard JDBC package, which allows you to select, insert, update, and delete data in SQL tables, add the following imports to your source code

```
import java.sql.* ; // for standard JDBC programs
import java.math.* ; // for BigDecimal and BigInteger support
```

Step 2: Load the drivers using the forName() method

- You must register the driver in your program before you use it. Registering the driver is the process by which the MySQL driver's class file is loaded into the memory, so it can be utilized as an implementation of the JDBC interfaces.
- You need to do this registration only once in your program. You can register a driver by using Java's Class.forName() method.
- The most common approach to register a driver is to use Java's Class.forName() method, to dynamically load the driver's class file into memory, which automatically registers it. This method is preferable because it allows you to make the driver registration configurable and portable.

RDBMS	JDBC driver name
MySQL	com.mysql.jdbc.Driver
ORACLE	oracle.jdbc.driver.OracleDriver
DB2	COM.ibm.db2.jdbc.net.DB2Driver
Sybase	com.sybase.jdbc.SybDriver

- The following example uses Class.forName() to register the MySQL driver

```
String driverName="com.mysql.cj.jdbc.Driver";
```

```

try
{
    Class.forName(driverName);
}

catch (Exception e)
{
    System.out.println("ClassNotFoundException");
}

```

Step 3: Establish a connection using the Connection class object

- After loading the driver, establish connections as shown below as follows:
 - Connection con = DriverManager.getConnection(url,user,password)
 - user: Username from which your SQL command prompt can be accessed.
 - password: password from which the SQL command prompt can be accessed.
 - con: It is a reference to the Connection interface.
 - Url: Uniform Resource Locator which is created as shown below:

```

String
dburl="jdbc:mysql://localhost:3306/databaseName";
String dbuser="root";
String dbpass="";

Connection con=null;
try {

    con=
DriverManager.getConnection(dburl,dbuser,dbpass);
    System.out.println("Connection Success");
}
catch (Exception e)
{
    System.out.println("Connection Failed::"+e);
}

```

URL format

```
jdbc:mysql://hostname:port/databaseName  
jdbc:oracle:thin:@hostname:port Number:databaseName  
jdbc:db2:hostname:port Number/databaseName  
jdbc:sybase:Tds:hostname: port Number/databaseName
```

Step 4: Create a statement

- Once a connection is established you can interact with the database. The JDBCStatement, CallableStatement, and PreparedStatement interfaces define the methods that enable you to send SQL commands and receive data from your database.
- Use of JDBC Statement is as follows:
 - **Statement st = con.createStatement();**
 - Note: Here, con is a reference to Connection interface used in previous step.
 - Note: In real-world application, it is recommended to use PreparedStatement or CallableStatement instead of Statement for executing SQL queries.
- The following table provides a summary of each interface's purpose to decide on the interface to use.

Interfaces	Recommended Use
Statement	Use this for general-purpose access to your database. Useful when you are using static SQL statements at runtime. The Statement interface cannot accept parameters.
PreparedStatement	Use this when you plan to use the SQL statements many times. The PreparedStatement interface accepts input parameters at runtime.
CallableStatement	Use this when you want to access the database stored procedures. The CallableStatement interface can also accept runtime input parameters.

Step 5: Execute the query

- Now comes the most important part i.e. executing the query. The query here is an SQL Query.
- Now, we know we can have multiple types of queries. Some of them are as follows:
 - The query for updating/inserting a table in a database.
 - The query for retrieving data.
 - The **executeQuery()** method of the Statement interface is used to execute queries of **retrieving values from the database**. This method returns the object of **ResultSet** that can be used to get all the records of a table.
 - **The executeUpdate(sql query)** method of the Statement interface is used to execute queries of **updating/inserting**.

```
Statement st=con.createStatement();

//Step:4:Write insert Query for faculty table.

String sql="query Statement";

//Step:5: Execute Query--->for
insert,delete,update use executeUpdate()
int r=st.executeUpdate(sql);
if (r>0)
{
    System.out.println();
    System.out.println("insertion Success");
    System.out.println();
}
else
{
    System.out.println("Insertion failed");
}
```

Step 6: Close the connections

- con.close();

Statement

- A Statement object is used for general-purpose access to databases and is useful for executing static SQL statements at runtime.
- Syntax:

Statement statement = connection.createStatement();

- Implementation: Once the Statement object is created, there are three ways to execute it.
 - **execute(String SQL):** It is used to executes any SQL statements (like SELECT, INSERT, UPDATE or DELETE). If the ResultSet object is retrieved, then it returns true else false is returned.
 - **executeUpdate(String SQL):** It is used to executes SQL statements (like INSERT, UPDATE or DELETE). It returns the number of rows affected by the SQL statement.
 - **ResultSet executeQuery(String SQL):** It is used to executes the SELECT query. It returns a ResultSet object that contains the data retrieved by the query.

Insert Data using Statement

```
import java.sql.*;
/**1**// To Establish Connection With database
public class App
{
    public static void main(String[] args) throws Exception
    {
        String dburl="jdbc:mysql://localhost:3306/lju";
        String dbuser="root";
        String dbpass="";
        String driverName="com.mysql.jdbc.Driver";

        //Step:1: Load Driver
```

```

        Class.forName(driverName);
        //Step:2:Connect with db

Connection con=DriverManager.getConnection(dburl,dbuser,dbpass);

        //check is connection Established?
        if(con!=null)
        {
            System.out.println("Connection Success");
        }
        else
        {
            System.out.println("Connection failed");
        }
    }
}

```

```

/**2**//InsertData
import java.sql.*;
class InsertJDBC
{
    public static void main(String[] args) throws Exception
    {
        String dburl="jdbc:mysql://localhost:3306/lju";
        String dbuser="root";
        String dbpass="";
        String driverName="com.mysql.cj.jdbc.Driver";
        //Step:1: Load Driver
        Class.forName(driverName);
        //Step:2:Connect with db
        Connection
con=DriverManager.getConnection(dburl,dbuser,dbpass);
        //check is connection Established?
        if(con!=null)
        {
            System.out.println();
            System.out.println("Connection Success");
            System.out.println();
        }
        else
        {
            System.out.println("Connection failed");
        }
        //Step:3:Create statement obj
    }
}

```

```

Statement st=con.createStatement();

//Step:4:Write insert Query for faculty table.

String sql="insert into Faculty(fid,fname,salary,number)
values(2,'Arp',25000,989898)";

//No primary key no autoincrement so we have to pass all the
values of table in query.
//if we are passing all the values of table then no need to
write attributes in query. It can be written as follows.

//String sql="insert into Facultyvalues(2,'Arp',25000,989898)";

//by making fid as primary key but no autoincrement so we have
to pass all the values of table in query but here we have
created fid as primary key so duplication is not allowed.

//by making fid as primary key and with autoincrement there
isn't any requirement to pass fid.

String sql="insert into Faculty(fname,salary,number)
values('Arp',25000,989898)";

//Step:5: Execute Query--->
//for insert,delete,update use executeUpdate()

int r=st.executeUpdate(sql);

if (r>0)
{
    System.out.println();
    System.out.println("insertion Success");
    System.out.println();
}
else
{
    System.out.println("Insertion failed");
}
}
}

```

```

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;
import java.util.Scanner;

/**3**//insertion using userInput
class InsertJDBC3
{
    public static void main(String[] args) throws Exception
    {
        String dburl="jdbc:mysql://localhost:3306/lju";
        String dbuser="root";
        String dbpass="";
        String driverName="com.mysql.cj.jdbc.Driver";

        //Step:1: Load Driver
        Class.forName(driverName);

        //Step:2:Connect with db

        Connection con= DriverManager.getConnection(dburl,dbuser,dbpass);

        //check is connection Establihed?

        if(con!=null)
        {
            System.out.println();
            System.out.println("Connection Success");
            System.out.println();
        }
        else
        {
            System.out.println("Connection failed");
        }

        //Step:3:Create statement obj

        Statement st=con.createStatement();

        Scanner sc=new Scanner(System.in);
        System.out.print("How many rows you want to enter::");
        int n=sc.nextInt();
        for(int i=1;i<=n;i++)

```

```

        {
            System.out.println("Enter Data for fac="+i);
            System.out.print("Enter Name::");
            String name=sc.next();
            System.out.print("Enter salary::");
            float sal=sc.nextFloat();
            System.out.print("Enter Number::");
            String num=sc.next();

            String sql="insert into faculty(facName,salary,number)
values ('"+name+"',"+sal+", "+num+" )";

            //Step:5: Execute Query--->for insert,delete,update
            use executeUpdate()

            int r=st.executeUpdate(sql);
            if (r>0)
            {
                System.out.println();
                System.out.println("insertion Success"+i);
                System.out.println();
            }
            else
            {
                System.out.println("Insertion failed"+i);
            }
        }
    }
}

```

Update Data using Statement

```

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.Statement;
import java.util.Scanner;

/**3**//insertion using userInput
class InsertJDBC3
{
    public static void main(String[] args) throws Exception
    {
        String dburl="jdbc:mysql://localhost:3306/lju";
        String dbuser="root";
        String dbpass="";
    }
}

```

```

String driverName="com.mysql.cj.jdbc.Driver";
Class.forName(driverName);

Connection con= DriverManager.getConnection(dburl,dbuser,dbpass);
if (con!=null)
{
    System.out.println();
    System.out.println("Connection Success");
    System.out.println();
}
else
{
    System.out.println("Connection failed");
}

Statement st=con.createStatement();
String sql="update faculty set salary=25000 where
facName='jjs'"; // query

//Different ways to write update query

/*String sql1="update faculty set number=979721 where facName='jjs'
AND salary=25000"; // query
String sql2="update faculty set number=212122, salary=50000 where
facName='vvp'"; // query
String sql3="update faculty set salary=75000";*/

int r=st.executeUpdate(sql);
if (r>0)
{
    System.out.println();
    System.out.println("insertion Success"+i);
    System.out.println();
}
else
{
    System.out.println("Insertion failed"+i);
}

}
}

```

Delete Data using Statement

```
import java.sql.*;
import java.util.*;
class deleteJDBC1
{
    public static void main(String[] args) throws Exception
    {
        String dburl="jdbc:mysql://localhost:3306/lju";
        String dbuser="root";
        String dbpass="";
        String driverName="com.mysql.cj.jdbc.Driver";

        Class.forName(driverName);

        Connection con=DriverManager.getConnection(dburl,dbuser,dbpass);

        if(con!=null)
        {
            System.out.println();
            System.out.println("Connection Success");
            System.out.println();
        }
        else
        {
            System.out.println("Connection failed");
        }

        Statement st=con.createStatement();

        String sql="delete from faculty where facName='jjs'"; // query
        String sql1="delete from faculty where salary=75000"; // query

        int r=st.executeUpdate(sql1);
        //returns int values no. of rows effected
        if (r>0)
        {
            System.out.println();
            System.out.println("Deletion Success in "+r+"
rows");
            System.out.println();
        }
        else
```

```

        {
            System.out.println("Deletion failed");
        }
    }
}

```

Retrive Data from Database using Statement

```

//Ex1
import java.sql.*;
import java.util.*;
public class selectJDBC1
{
    public static void main(String[] args) throws Exception
    {
        String dburl="jdbc:mysql://localhost:3306/lju";
        String dbuser="root";
        String dbpass="";
        String driverName="com.mysql.cj.jdbc.Driver";

        Class.forName(driverName);

        Connection con=DriverManager.getConnection(dburl,dbuser,dbpass);

        if(con!=null)
        {
            System.out.println();
            System.out.println("Connection Success");
            System.out.println();
        }
        else
        {
            System.out.println("Connection failed");
        }

        Statement st=con.createStatement();

        String sql="select * from faculty";
        // to retrieve all data

        ResultSet rs=st.executeQuery(sql);
        while(rs.next()) //returns true false
        {

```

```

        //display data on consol
        System.out.println("FacId="+rs.getInt(1));
        System.out.println("FacName="+rs.getString(2));
        System.out.println("Salary="+rs.getFloat("salary"));

        System.out.println("number="+rs.getString("number"));
    }
}
}

```

```

//Ex2
import java.sql.*;
import java.util.*;
class selectJDBC2
{
    public static void main(String[] args) throws Exception
    {
        String dburl="jdbc:mysql://localhost:3306/lju";
        String dbuser="root";
        String dbpass="";
        String driverName="com.mysql.cj.jdbc.Driver";

        Class.forName(driverName);

        Connection con=DriverManager.getConnection(dburl,dbuser,dbpass);

        if(con!=null)
        {
            System.out.println();
            System.out.println("Connection Success");
            System.out.println();
        }
        else
        {
            System.out.println("Connection failed");
        }

        Statement st=con.createStatement();

        String sql="select facName,salary from faculty where number='12345'";

        ResultSet rs=st.executeQuery(sql);

        while(rs.next()) //returns true false
        {

```

```

        //Display data on console
        System.out.println("FacName="+rs.getString(1));
        System.out.println("Salary="+rs.getFloat(2));
    }
}
}

```

/9**// CRUD**

```

import java.sql.*;
import java.util.*;
class crudJDBC1
{
    static Connection con=null;
    static Statement st=null;
    static Scanner sc=new Scanner(System.in);
    static void insertData() throws Exception
    {
        System.out.print("Enter Project Name::");
        String pname=sc.next();
        System.out.print("Enter Project Type::");
        String ptype=sc.next();
        System.out.print("Enter Project Cost::");
        float pcost=sc.nextFloat();

        String sql="insert into project (pname,ptype,pcost)
values('"+pname+"','"+ptype+"','"+pcost+"')";
        int r=st.executeUpdate(sql);
        if (r>0)
        {
            System.out.println();
            System.out.println("Insert Success in "+r+" rows");
            System.out.println();
        }
        else
        {
            System.out.println("Insert failed");
        }
    }
    static void deleteData() throws Exception
    {
        System.out.print("Enter project name you want to
delete::");
        String name=sc.next();
    }
}

```

```

        String sql= "delete from project where
pname='"+name+"'";
        int r=st.executeUpdate(sql);
        if (r>0)
        {
            System.out.println();
            System.out.println("Deletion Success in "+r+"
rows");
            System.out.println();
        }
        else
        {
            System.out.println("Deletion failed");
        }
    }
    static void updateData() throws Exception
    {
        System.out.print("Enter project's cost you want to
update::");
        String name=sc.next();
        System.out.print("Enter New Cost");
        float cost=sc.nextFloat();
        String sql="update Project set pcost="+cost+" where
pname='"+name+"'";
        int r=st.executeUpdate(sql);
        if (r>0)
        {
            System.out.println();
            System.out.println("Update Success in "+r+" rows");
            System.out.println();
        }
        else
        {
            System.out.println("Update failed");
        }
    }
    static void showWebsite() throws Exception
    {
        String sql="select * from project where
ptype='website'"; // query

        ResultSet rs=st.executeQuery(sql);//returns int values
no. of rows effected
        while(rs.next()) //returns true false
        {
            System.out.println();
            System.out.println("projectId="+rs.getInt(1));

```

```

        System.out.println("projectName="+rs.getString(2));
        System.out.println("project type="+rs.getString(3));
        System.out.println("project cost="+rs.getFloat(4));
        System.out.println();
    }
}
static void showApp() throws Exception
{
    String sql="select * from project where ptype='APP'; //
query
    ResultSet rs=st.executeQuery(sql);//returns int values
no. of rows effected
    while(rs.next()) //returns true false
    {
        System.out.println();
        System.out.println("projectId="+rs.getInt(1));
        System.out.println("projectName="+rs.getString(2));
        System.out.println("project type="+rs.getString(3));
        System.out.println("project cost="+rs.getFloat(4));
        System.out.println();
    }
}
static void showAll() throws Exception
{
    String sql="select * from project"; // query
    ResultSet rs=st.executeQuery(sql);//returns int values
no. of rows effected
    while(rs.next()) //returns true false
    {
        System.out.println();
        System.out.println("projectId="+rs.getInt(1));
        System.out.println("projectName="+rs.getString(2));
        System.out.println("project type="+rs.getString(3));
        System.out.println("project cost="+rs.getFloat(4));
        System.out.println();
    }
}
public static void main(String[] args)throws Exception
{
    String dburl="jdbc:mysql://localhost:3306/lju";
    String dbuser="root";
    String dbpass="";
    String driverName="com.mysql.cj.jdbc.Driver";

    Class.forName(driverName);

```

```

con=DriverManager.getConnection(dburl,dbuser,dbpass);
if(con!=null)
{
    System.out.println();
    System.out.println("Connection Success");
    System.out.println();
}
else
{
    System.out.println("Connection failed");
}
st=con.createStatement();
int ch;
do
{
    System.out.print("Enter 1 for Insert \n Enter 2 for
Delete \n Enter 3 for Update \n Enter 4 for Select \n Enter 5
for Exit \n");
    System.out.print("Enter your Choice =");
    ch=sc.nextInt();
    switch(ch)
    {
        case 1:
            insertData();
            break;
        case 2:
            deleteData();
            break;
        case 3:
            updateData();
            break;
        case 4:
            int ch1;
            do
            {
                System.out.println();
                System.out.print("Enter 1 for view
WebsiteData \n Enter 2 for APP data \n Enter 3 for ALL \n Enter
4 to go back \n");
                System.out.print("Enter your
Choice =");
                ch1=sc.nextInt();
                switch(ch1)
                {
                    case 1:
                        showWebsite();
                        break;

```

```

        case 2:
            showApp();
            break;
        case 3:
            showAll();
            break;
        case 4:
            System.out.println();
            break;
        default:
            System.out.println();
            System.out.println("*****enter
valid Choice*****");
    }
    }while(ch1!=4);
    case 5:
        break;
    default:
        System.out.println();
        System.out.println("*****enter valid
choice*****");
        System.out.println();
    }
    }while(ch!=5);
}
}

```